



NORTH SYDNEY BOYS HIGH SCHOOL

2020 YEAR 12 ASSESSMENT TASK 3

Mathematics Advanced

General Instructions

- Working time – 3 hours
- **Reading time – 10 minutes**
- Write on the lined paper in the booklet provided
- Write Multiple Choice responses on sheet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

- Use the Reference Sheet

Class Teacher: Please shade the circle.

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Ms Lee
- ☐ Mr Umakanthan

STUDENT NUMBER: 4.....

(To be used by the exam markers only.)

Questions	1-10	11-17	18-21	22-25	26-29	30-34	35-38	Total
Mark	$\overline{10}$	$\overline{15}$	$\overline{16}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{14}$	$\overline{100}$

Section I

Use the multiple-choice answer sheet for Questions 1 – 10

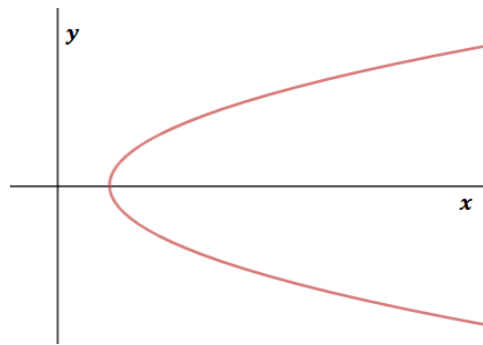
1. The period of $f(x) = 4 \cos\left(\frac{2\pi x}{3}\right) + 1$ is:

- A. $\frac{2}{3}$
- B. $\frac{3}{2}$
- C. 3
- D. 5

2. If $f(x) = \frac{x^2+1}{4-x^2}$ then the vertical and horizontal asymptotes are respectively:

- A. $x = -2, \quad x = 2, \quad y = \frac{1}{4}$
- B. $x = -2, \quad x = 2, \quad y = -1$
- C. $x = -2, \quad x = 2, \quad y = 1$
- D. $x = -2, \quad x = 2, \quad y = 0$

3. The graph below shows the relation $y^2 = x - 1$. What type of relation is it?



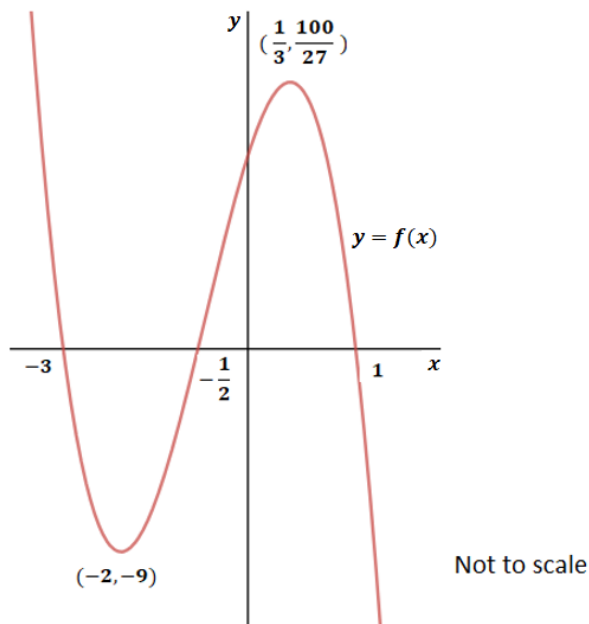
- A. one-to-one
- B. many-to-one
- C. many-to-many
- D. one-to-many

4. The 7th term of an arithmetic sequence is 45 and the 11th term is 77.

Find the first term (a) and the common difference (d).

- A. $a = -3$ and $d = 8$
- B. $a = 3$ and $d = 8$
- C. $a = 8$ and $d = -3$
- D. $a = 8$ and $d = 3$

5. Part of the graph $y = f(x)$ of the function f is shown:



$f'(x) < 0$ for

- A. $x \in (-3, -\frac{1}{2}) \cup (1, \infty)$
- B. $x \in (-9, \frac{100}{27})$
- C. $x \in (-2, \frac{1}{3})$
- D. $x \in (-\infty, -2) \cup (\frac{1}{3}, \infty)$

6. Find the derivative of $e^{x \sin 3x}$.

- A. $e^{3x \cos 3x}$
- B. $e^{x \sin 3x}(\sin 3x + 3x \cos 3x)$
- C. $e^{x \sin 3x}$
- D. $e^{x \sin 3x}(\sin 3x - 3x \cos 3x)$

7. The graph of a function f is obtained from the graph of the function g with rule $g(x) = \sqrt{2x - 5}$ by a reflection in the x -axis, followed by a horizontal dilation from the y -axis by a factor of $\frac{1}{2}$.

Which one of the following is the rule for the function f ?

- A. $f(x) = \sqrt{5 - 4x}$
- B. $f(x) = -\sqrt{x - 5}$
- C. $f(x) = \sqrt{x + 5}$
- D. $f(x) = -\sqrt{4x - 5}$

8. The discrete random variable X has this probability distribution:

x	0	1	2	3
$P(X=x)$	a	$3a$	$5a$	$7a$

The mean of X is:

- A. $\frac{1}{16}$
- B. 1
- C. $\frac{35}{16}$
- D. $\frac{17}{8}$

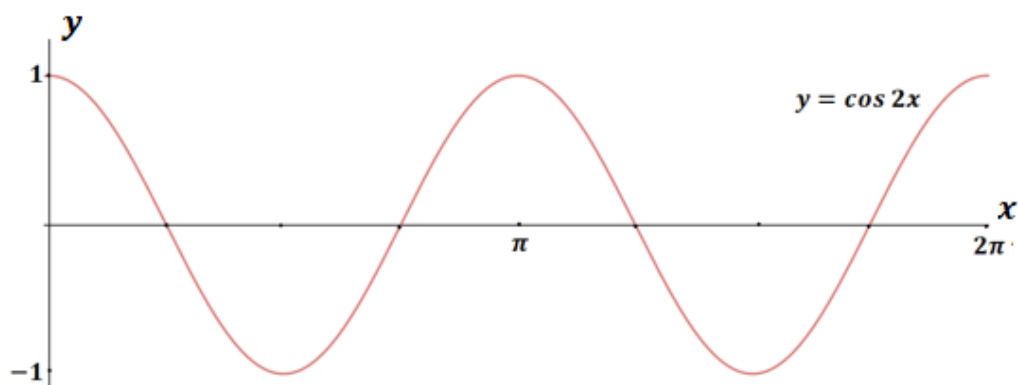
9. The difference in intensity of two sources of sound P_1 and P_2 is defined

to be $10 \log_{10} \left(\frac{P_1}{P_2} \right)$ decibels. How much louder is a sound of 112 dB than a sound of 80 dB ?

- A. 32 times
- B. 1 585 times
- C. 25 times
- D. 3.2 times

10. How many solutions does $6 \cos 2x = x$ for $0 \leq x \leq 2\pi$ have ?

(note: graph below shows $y = \cos 2x$)



- A. 1
- B. 2
- C. 3
- D. 4

Section II: Short Answer

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet. If you use this space, clearly indicate which question you are answering.

Question 11 (1 mark)

Factorise $2x^2 + 5x + 2$

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Question 12 (2 marks)

Rationalise the denominator: $\frac{2}{3-\sqrt{2}}$

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Question 13 (3 marks)

Find the following integrals:

(a) $\int \frac{1-2x^5}{x} dx$

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(b) $\int (3x + 2)^4 dx$

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Question 14 (3 marks)

Differentiate the following functions:

(a) $y = 3^{5x+2}$

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(b) $y = \frac{x}{\log_e x}$

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Question 15 (1 mark)

Write down the domain of $g(x) = \log_e(x + \pi)$

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Question 16 (3 marks)

Solve $2 \log_e x = \log_e(2x + 3)$

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Question 17 (2 marks)

Solve the inequality $x^2 \geq 3x + 18$

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Question 18 (2 marks)

Find $\int_1^{e^3} \frac{5}{x} dx$

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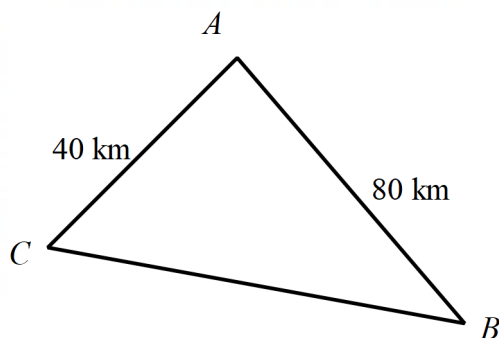
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Question 19 (4 marks)

Three towns, A , B and C form a triangle.

Town A is 80 km from Town B and Town C is 40 km from Town A as shown below:



The bearing of Town B from Town A is 130° . The bearing of Town C from Town A is 240°

- (a) Use this information to find the size of $\angle CAB$, and hence find the area of the triangle formed by the three towns to the nearest square kilometre.

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- (b) Using the cosine rule, find the distance between Town B and Town C , to the nearest kilometre.

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Question 20 (3 marks)

- (a) Given $f(x) = \sqrt{4 - x^2}$ complete this table of values, correct to 3 decimal places. 1

x	0	0.5	1	1.5	2
$f(x)$					

- (b) Use the Trapezoidal rule, with four sub-intervals, to estimate the value of

$$\int_0^2 \sqrt{4 - x^2} dx.$$

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Question 21 (7 marks)

For the curve: $y = x^3 - 3x^2 - 9x + 4$

(a) Find any stationary points and determine their nature.

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(b) Find any points of inflexion.

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(c) Sketch the curve, showing all main features.

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Question 22 (2 marks)

Find the exact value of $\cot \theta$ given that $\cos \theta = 0.6$ and $\sin \theta < 0$.

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Question 23 (3 marks)

A geometric progression has 5th term 9 and 13th term 59 049.

(a) Find the first term and the common ratio.

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(b) Find the 19th term.

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Question 24 (5 marks)

The number of bacteria in a culture can be modelled by $B = 120\,000 e^{0.4 t}$
where t is the time in hours after the experiment started.

- (a) How many bacteria are there after 6 hours have passed? **1**

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- (b) How fast was the culture growing after 6 hours? **1**

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- (c) What was the average rate of increase over the first 6 hours? **1**

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- (d) How long, in hours and minutes, will it take until the number of
bacteria doubles? **2**

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Question 25 (5 marks)

In an experiment, 2 balls are drawn at random and without replacement from an urn containing 4 red balls and 6 black balls. Let X be the number of red balls selected.

(a) Complete the table below:

2

Outcome	RR	RB	BR	BB	
X	2	1	1	0	
$p(X = x)$	$\frac{2}{15}$				
$x \cdot p(x)$					
X^2					

(b) What is the expected number of red balls drawn?

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(c) What is the variance, $V(X)$, of this distribution?

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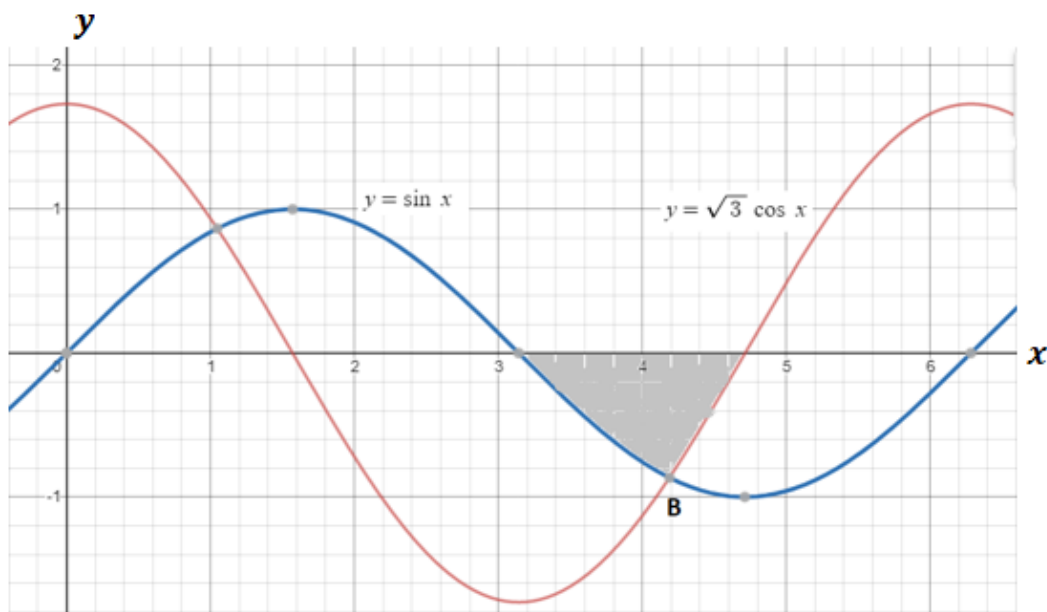
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Question 26 (4 marks)



The diagram above shows the graphs of $y = \sin x$ and $y = \sqrt{3} \cos x$, $0 \leq x \leq 2\pi$.
The second point of intersection is labelled B .

- (a) Show, using any appropriate method, that B has coordinates $\left(\frac{4\pi}{3}, \frac{-\sqrt{3}}{2}\right)$

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- (b) Find the exact area of the shaded region.

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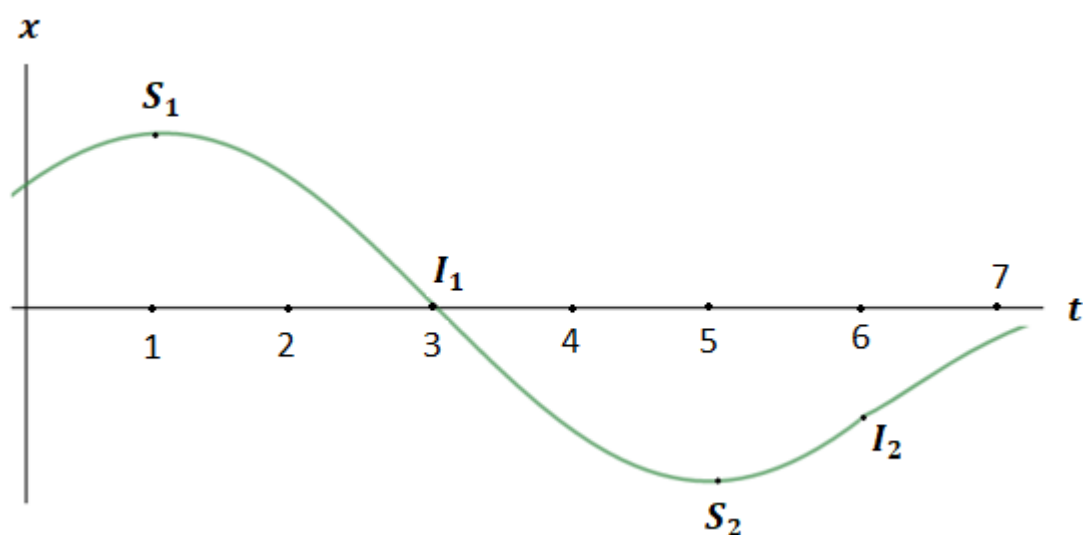
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Question 27 (3 marks)



The graph shows the displacement of a particle, moving in a straight line, over the first 7 seconds of its motion. S_1 and S_2 are stationary points, and I_1 and I_2 are inflection points.

State the times, or periods of time, for which:

(a) The particle is stationary.

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(b) The velocity is negative.

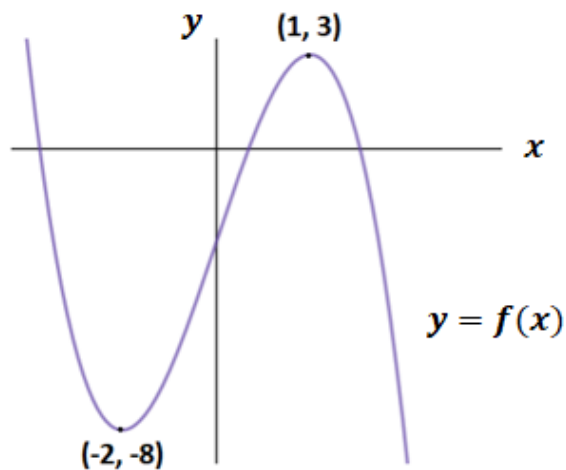
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(c) The acceleration is positive.

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Question 28 (4 marks)

Consider the graph of $y = f(x)$ shown:



- (a) Use the space below to sketch the graph of $y = f'(x)$

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- (b) Find the area bounded by $y = f'(x)$ and the x-axis.

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Question 29 (4 marks)

For events A and B from a sample space, $P(A|B) = \frac{3}{4}$ and $P(B) = \frac{1}{3}$

(a) Calculate $P(A \cap B)$

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(b) Calculate $P(\bar{A} \cap B)$ where $\bar{A}$ denotes the complement of A.

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(c) If A and B are independent, calculate $P(A \cup B)$

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Question 30 (2 marks)

The gradient of a curve is given by $\frac{dy}{dx} = \frac{3x}{x^2 + e}$

The curve passes through (0, 2). What is its equation?

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Question 31 (3 marks)

If $f(x) = \sqrt{2 - x}$ and $g(x) = \sqrt{x}$, then

(a) Find the rule for the composite function $f \circ g$

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(b) Find the domain of $f \circ g$

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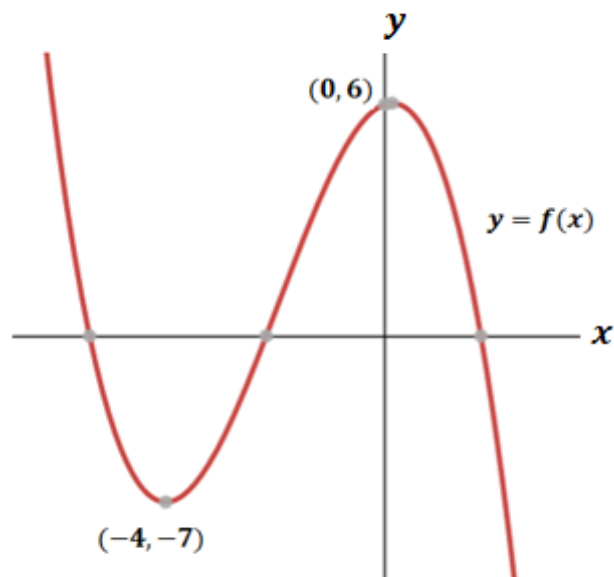
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Question 32 (2 marks)

Given the graph of the function $y = f(x)$ below, with turning points as shown, sketch the transformed function $y = 3f(x + 2) - 4$. (x-intercepts not required).

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Question 33 (5 marks)

- (a) Differentiate $y = \log_e (\cos x)$ with respect to x . **1**

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- (b) Sketch $y = \tan x$ for $0 \leq x \leq \frac{\pi}{2}$ **1**

- (c) Hence, using parts (a) and (b), find the area bounded by $y = \tan x$,
the x -axis, and the line $x = \frac{\pi}{3}$ **3**
(leave answer in simplest exact form)

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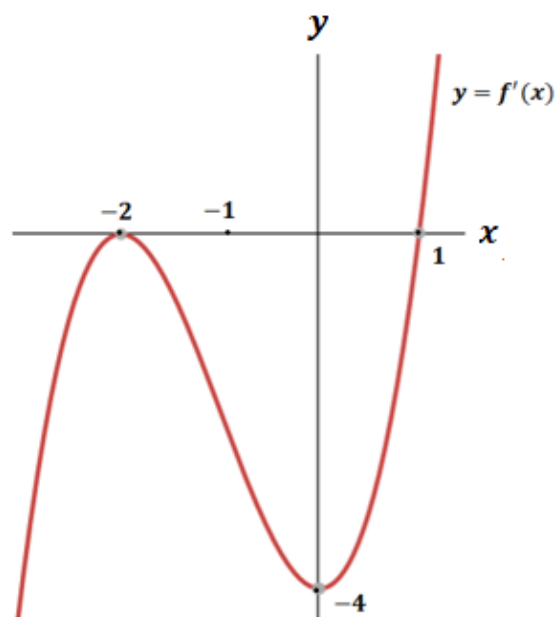
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Question 34 (3 marks)

The diagram shows $y = f'(x)$, the graph of the derivative function of $y = f(x)$.



- (a) Explain why there is a horizontal point of inflection at $x = -2$

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- (b) Given that $f(0) = 2$, sketch a possible graph of $y = f(x)$.

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Question 35 (3 marks)

Find the equation of the normal to $y = x \sin x$ at the point where $x = \frac{\pi}{2}$.

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Question 36 (2 marks)

(a) Sketch $y = |x| - 1$ and $y = 2x + 2$ neatly on the same number plane.

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(b) Hence solve the equation $|x| - 2x = 3$

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Question 37 (4 marks)

One half percent (0.5 %) of a country has a certain viral disease. A test is developed for the disease. The test gives a false positive 3% of the time, and a false negative 2% of the time.

- (a) Show that the probability that Andy, a randomly selected person, tests positive is 0.03475 2

[Hint: in this question, let D be the event that Andy has the disease, and $\bar{D}$ be the event Andy does not have it. Let T be the event that Andy's test comes back positive.]

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- (b) Andy just got the bad news that his test came back positive.
Find the probability that Andy actually has the disease.2

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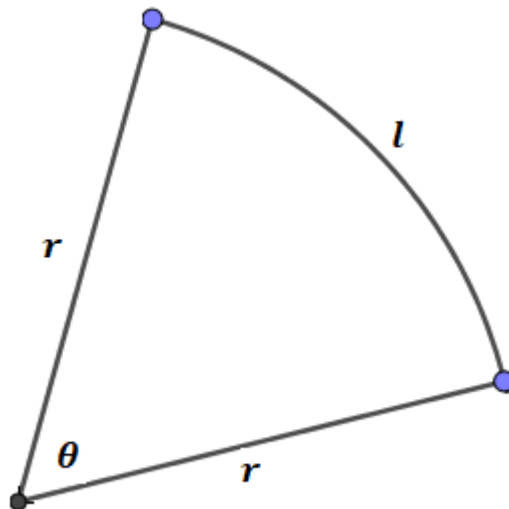
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Question 38 (5 marks)

The diagram below shows a sector of a circle of radius r centimetres. The angle at the centre is θ radians, and the perimeter of the whole sector is 8 cm .



- (a) Show that $r = \frac{8}{2+\theta}$.

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- (b) Show that A , the area of the sector in cm^2 , is given by

$$A = \frac{32\theta}{(\theta+2)^2}$$

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This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines, providing a guide for letter height and placement. The lines are evenly spaced across the entire page, leaving ample room for writing practice. There is no text or other markings on the page.

26

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g "●"

STUDENT NUMBER:

Class (please ✓)

☐ - Mr Ireland

☐ - Dr Jomaa

☐ - Miss Lee

☐ - Mr Lin

☐ - Mrs Sarofim

☐ - Ms Ziazaris

☐ - Mr Berry

☐ - Mr Hwang

☐ - Mr Uma

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|------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1 - | ☐ A | ☐ B | ☒ C | ☐ D |
| 2 - | ☐ A | ☒ B | ☐ C | ☐ D |
| 3 - | ☐ A | ☐ B | ☐ C | ☒ D |
| 4 - | ☒ A | ☐ B | ☐ C | ☐ D |
| 5 - | ☐ A | ☐ B | ☐ C | ☒ D |
| 6 - | ☐ A | ☒ B | ☐ C | ☐ D |
| 7 - | ☐ A | ☐ B | ☐ C | ☒ D |
| 8 - | ☐ A | ☐ B | ☐ C | ☒ D |
| 9 - | ☐ A | ☒ B | ☐ C | ☐ D |
| 10 - | ☐ A | ☐ B | ☒ C | ☐ D |

2020 Mathematics Advanced - Task 3

$$(1) \quad T = \frac{2\pi}{\frac{2\pi}{3}} = 3 \quad \therefore (C)$$

$$(2) \quad f(x) = \frac{x^2 + 1}{4 - x^2} \quad \therefore x \neq \pm 2 \quad \therefore (B)$$

as $x \rightarrow \infty$, $y \rightarrow -1$

$$(3) \quad \text{one-to-many,} \quad \therefore (D)$$

$$(4) \quad \begin{aligned} T_7 &= a + 6d = 45 \\ T_{11} &= a + 10d = 77 \\ \therefore 4d &= 32, \quad d = 8 \\ \therefore a &= -3 \end{aligned} \quad \therefore (A)$$

$$(5) \quad f'(x) < 0 \text{ means gradient is negative, } \therefore x < -2 \quad \therefore (D)$$

or $x > \frac{1}{3}$

$$(6) \quad \frac{d}{dx} e^{x \sin 3x} = (\sin 3x \cdot 1 + x \cdot 3 \cos 3x) \cdot e^{x \sin 3x} \quad \therefore (B)$$

$$(7) \quad \sqrt{2x-5} \rightarrow -\sqrt{2x-5} \rightarrow -\sqrt{2\left(\frac{x}{2}\right)-5} \quad \therefore (D)$$

$= -\sqrt{4x-5}$

⑧ $\sum p = 1 \therefore a + 3a + 5a + 7a = 1$
 $\therefore a = \frac{1}{16}$

$$\begin{aligned}\mu &= \sum x \cdot p(x) = 0 + 3a + 10a + 21a \\ &= 34a \\ &= \frac{34}{16} = \frac{17}{8}\end{aligned}$$

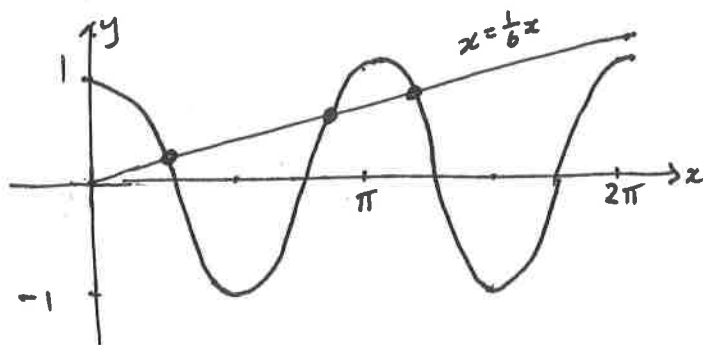
(9) $112 \text{ dB} - 80 \text{ dB} = 32 \text{ dB}$
 $\therefore \log_{10}\left(\frac{P_1}{P_2}\right) = 3.2 \quad \therefore \frac{P_1}{P_2} = 10^{3.2} \quad \therefore \text{(B)}$
 $= 1585$

(10) $6 \cos 2x = x$
means $\cos 2x = \frac{1}{6}x$.

If we draw $y = \frac{1}{6}x$, then at

$$x = 2\pi, \quad y \doteq 1.047 > \cos 2(2\pi)$$

$\therefore$ 3 intersections from $x=0$ to $x=2\pi$



Section II: Short Answer

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet.
If you use this space, clearly indicate which question you are answering.

Question 11 (1 mark)

Factorise $2x^2 + 5x + 2$

1

$$(2x+1)(x+2)$$

Question 12 (2 marks)

Rationalise the denominator: $\frac{2}{3-\sqrt{2}}$

2

$$= \frac{2}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}}$$

$$= \frac{6 + 2\sqrt{2}}{9-2}$$

$$= \frac{6 + 2\sqrt{2}}{7}$$

Question 13 (3 marks)

Find the following integrals:

(a) $\int \frac{1-2x^5}{x} dx$

2

$$= \int \left(\frac{1}{x} - 2x^4 \right) dx$$

$$= \ln x - \frac{2x^5}{5} + C$$

(b) $\int (3x+2)^4 dx$

1

=

$$\frac{(3x+2)^5}{5} + C$$

✓

Question 14 (3 marks)

Differentiate the following functions:

(a) $y = 3^{5x+2}$

1

$$y' = \ln 3 \cdot 5 \cdot 3^{5x+2}$$

i.e. $y' = 5 \ln 3 \cdot 3^{5x+2}$

✓

(b) $y = \frac{x}{\log_e x}$

2

$$y' = \frac{\log x \cdot 1 - x \cdot \frac{1}{x}}{(\log x)^2}$$

✓

$$\therefore y' = \frac{\log_e x - 1}{(\log_e x)^2}$$

✓

Question 15 (1 mark)

Write down the domain of $g(x) = \log_e(x + \pi)$

1

We need $x + \pi > 0 \therefore x > -\pi$

$$\therefore \mathcal{D} : x > -\pi$$

✓

Question 16 (3 marks)Solve $2 \log_e x = \log_e(2x+3)$

3

$$\log_e x^2 = \log_e(2x+3)$$

$$\therefore x^2 = 2x+3$$

$$\therefore x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$\therefore x = 3 \text{ or } x = -1$$

But $x \neq -1$ as this can't be subbed into the equation

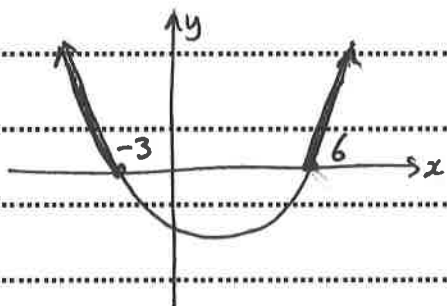
$$\therefore \boxed{x = 3}$$

Question 17 (2 marks)Solve the inequality $x^2 \geq 3x + 18$

2

$$x^2 - 3x - 18 \geq 0$$

$$(x-6)(x+3) \geq 0$$



$$\therefore \boxed{x \leq -3 \text{ or } x \geq 6}$$

Question 18 (2 marks)Find $\int_1^{e^3} \frac{5}{x} dx$

2

$$= [5 \ln x]_1^{e^3}$$

$$= 5 \ln e^3 - 5 \ln 1$$

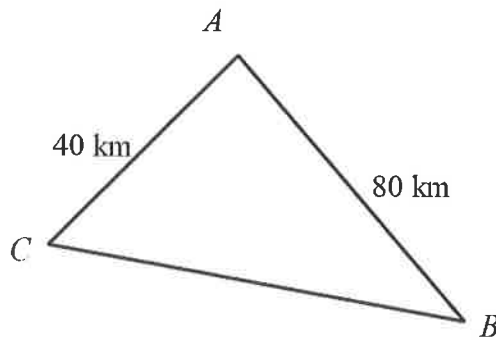
$$= 5 \times 3 - 5 \times 0$$

$$= \boxed{15}$$

Question 19 (4 marks)

Three towns, A , B and C form a triangle.

Town A is 80 km from Town B and Town C is 40 km from Town A as shown below:



The bearing of Town B from Town A is 130° . The bearing of Town C from Town A is 240°

- (a) Use this information to find the size of $\angle CAB$, and hence find the area of the triangle formed by the three towns to the nearest square kilometre.

2

$$\angle CAB = 240^\circ - 130^\circ = 110^\circ$$

$$\therefore \text{Area } \triangle ABC = \frac{1}{2} \cdot 40 \cdot 80 \cdot \sin 110^\circ$$

$$\doteq 1503.508193 \dots$$

$$= 1504 \text{ km}^2 \text{ (nearest km}^2\text{)}$$

✓✓

- (b) Using the cosine rule, find the distance between Town B and Town C , to the nearest kilometre.

2

$$BC^2 = 40^2 + 80^2 - 2(40)(80) \cos 110^\circ$$

✓

$$\doteq 10188.92892 \dots$$

$$\therefore BC \doteq 100.940224 \dots$$

$$\therefore BC = 101 \text{ km (nearest km)}$$

✓

Question 20 (3 marks)

(a) Given $f(x) = \sqrt{4-x^2}$ complete this table of values, correct to 3 decimal places.

1

x	0	0.5	1	1.5	2
$f(x)$	2	1.936	1.732	1.323	0



(b) Use the Trapezoidal rule, with four sub-intervals, to estimate the value of

2

$$\int_0^2 \sqrt{4-x^2} dx.$$

$$\int_0^2 \sqrt{4-x^2} dx$$

$h = \text{width sub-interval} = 0.5$

$$\div \frac{0.5}{2}$$

$$\left[2 + 0 + 2(1.936 + 1.732 + 1.323) \right]$$



$$= 2.9955$$



Question 21 (7 marks)

For the curve: $y = x^3 - 3x^2 - 9x + 4$

(a) Find any stationary points and determine their nature.

3

$$y' = 3x^2 - 6x - 9 \quad ; \quad y'' = 6x - 6$$
$$= 3(x^2 - 2x - 3) \quad \therefore y'' = 6(x-1)$$

✓

$$\therefore y' = 3(x-3)(x+1)$$

For stat. pts, $y' = 0 \therefore x = 3 \quad \text{or} \quad x = -1$

$$y = -23 \quad \quad \quad y = 9$$

✓

At $(3, -23)$, $y'' = 6(3-1) = 12 > 0 \therefore$ local minimum
at $(3, -23)$

At $(-1, 9)$, $y'' = 6(-1-1) = -12 < 0 \therefore$ local maximum
at $(-1, 9)$

✓

(b) Find any points of inflexion.

2

For inflexions, $y'' = 0$

$$\therefore 6(x-1) = 0 \therefore x = 1, y = -7.$$

✓

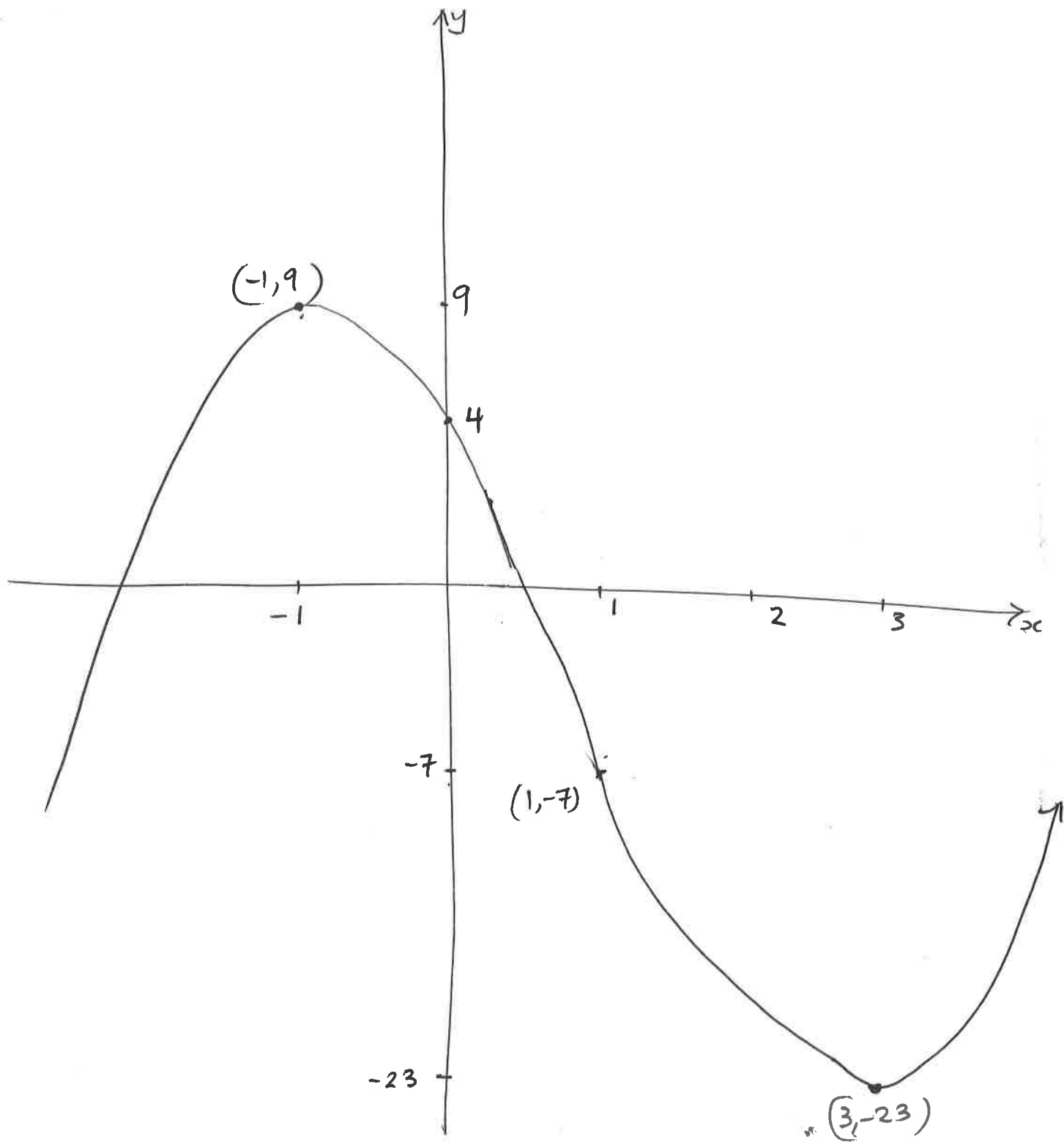
Test :

x	0	1	2
y''	-6	0	6

change in concavity, $\therefore$ inflexion at $(1, -7)$ ✓

(c) Sketch the curve, showing all main features.

2



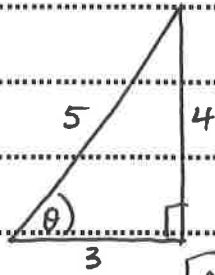
✓✓

Question 22 (2 marks)Find the exact value of $\cot \theta$ given that $\cos \theta = 0.6$ and $\sin \theta < 0$.

2

$$\cos \theta = 0.6 = \frac{3}{5}; \quad \cos \theta > 0 \text{ \& \; } \sin \theta < 0 \therefore Q4$$

$$\therefore \tan \theta < 0.$$



$$\therefore \tan \theta = -\frac{4}{3}$$

$$\therefore \cot \theta = -\frac{3}{4}$$

✓✓

$$[\text{ALT: } \sin \theta = -\sqrt{1 - \cos^2 \theta}, \text{ as in Q4} = -\sqrt{1 - \frac{9}{25}}$$

$$\therefore \sin \theta = -\frac{4}{5} \therefore \cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{\frac{3}{5}}{-\frac{4}{5}} = -\frac{3}{4}]$$

Question 23 (3 marks)A geometric progression has 5th term 9 and 13th term 59 049.

(a) Find the first term and the common ratio.

2

$$T_5 = ar^4 = 9$$

$$T_{13} = ar^{12} = 59049$$

$$\therefore \frac{ar^{12}}{ar^4} = \frac{59049}{9}$$

$$\therefore r^8 = 6561$$

$$\therefore r = 3 \text{ or } -3$$

$$\therefore a(81) = 9 \therefore a = \frac{1}{9}$$

✓✓

(b) Find the 19th term.

1

$$T_{19} = ar^{18} = \frac{1}{9} \cdot (3)^{18} \quad \left(\text{or } \frac{1}{9} (-3)^{18} \right)$$

$$= 3^{16}$$

$$= 43\,046\,721$$

✓

Question 24 (5 marks)

The number of bacteria in a culture can be modelled by $B = 120\,000 e^{0.4t}$

where t is the time in hours after the experiment started.

- (a) How many bacteria are there after 6 hours have passed?

1

$$B = 120\,000 e^{(0.4)6} = 120\,000 e^{2.4}$$

$$= 1\,322\,781$$

- (b) How fast was the culture growing after 6 hours?

1

$$\frac{dB}{dt} = 0.4 \times 120\,000 e^{0.4t} = 48\,000 e^{0.4t}$$

$$\therefore \text{at } t=6, \frac{dB}{dt} = 48\,000 e^{2.4}$$

$$= 529\,112 \text{ per hour.}$$

- (c) What was the average rate of increase over the first 6 hours?

1

$$\text{Starting number at } t=0 = 120\,000 e^0 = 120\,000$$

$$\therefore \text{average rate} = \frac{1\,322\,781 - 120\,000}{6}$$

$$= 200\,463 \text{ per hour}$$

- (d) How long, in hours and minutes, will it take until the number of bacteria doubles?

2

$$240\,000 = 120\,000 e^{0.4t}$$

$$\therefore e^{0.4t} = 2$$

$$\therefore 0.4t = \ln 2$$

$$\therefore t = \frac{\ln 2}{0.4} \doteq 1.732\,867... \text{ hours}$$

$$= 1 \text{ hour } 44 \text{ mins. (nearest min.)}$$

Question 25 (5 marks)

In an experiment, 2 balls are drawn at random and without replacement from an urn containing 4 red balls and 6 black balls. Let X be the number of red balls selected.

(a) Complete the table below:

2

Outcome	RR	RB	BR	BB	
X	2	1	1	0	
$p(X = x)$	$\frac{2}{15}$	$\frac{4}{10} \cdot \frac{6}{9} = \frac{4}{15}$	$\frac{6}{10} \cdot \frac{4}{9} = \frac{4}{15}$	$\frac{6}{10} \cdot \frac{5}{9} = \frac{1}{3}$	
$x \cdot p(x)$	$\frac{4}{15}$	$\frac{4}{15}$	$\frac{4}{15}$	4 0	
X^2	4	1	1	0	

✓✓

(b) What is the expected number of red balls drawn?

1

$$E(X) = \sum x \cdot p(x)$$

$$= \frac{4}{15} + \frac{4}{15} + \frac{4}{15} + 0 = \frac{12}{15}$$

$$= 0.8$$

✓

(c) What is the variance, $V(X)$, of this distribution?

2

$$E(X^2) = 4 \cdot \frac{2}{15} + 1 \cdot \frac{4}{15} + 1 \cdot \frac{4}{15} + 0 \cdot \frac{1}{3} = \frac{16}{15}$$

$$\therefore V(X) = E(X^2) - (E(X))^2$$

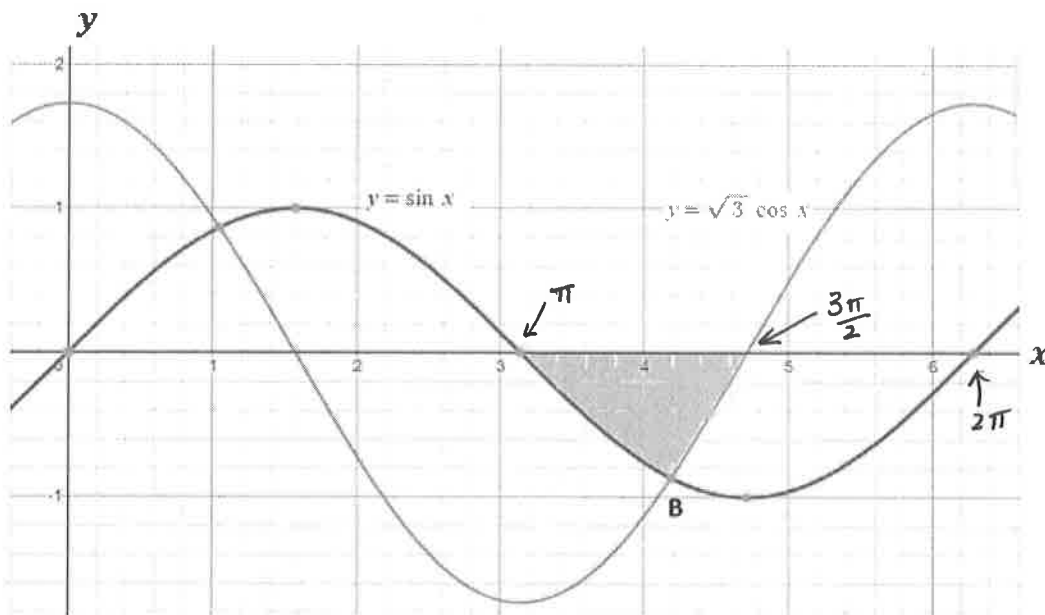
$$= \frac{16}{15} - (0.8)^2$$

$$= \frac{32}{75}$$

$$(= 0.427 \text{ 3d.p.})$$

✓✓

Question 26 (4 marks)



The diagram above shows the graphs of $y = \sin x$ and $y = \sqrt{3} \cos x$, $0 \leq x \leq 2\pi$. The second point of intersection is labelled B.

- (a) Show, using any appropriate method, that B has coordinates $(\frac{4\pi}{3}, -\frac{\sqrt{3}}{2})$ 1

B lies on $\sin x$ and on $\sqrt{3} \cos x$.
 Here, $\sin \frac{4\pi}{3} = -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2}$
 and $\sqrt{3} \cos \frac{4\pi}{3} = \sqrt{3} \cdot -\cos \frac{\pi}{3}$
 $= \sqrt{3} \cdot -\frac{1}{2} = -\frac{\sqrt{3}}{2}$
 $\therefore$ curves both go thru $(\frac{4\pi}{3}, -\frac{\sqrt{3}}{2})$

ALT: $\sin x = \sqrt{3} \cos x$
 $\therefore \tan x = \sqrt{3}$
 $\therefore x = \frac{\pi}{3}, \frac{4\pi}{3}$
 But $x > \pi \therefore x = \frac{4\pi}{3}$
 $\therefore y = \sin \frac{4\pi}{3} = -\frac{\sqrt{3}}{2}$

- (b) Find the exact area of the shaded region. 3

Area = $\left| \int_{\pi}^{\frac{4\pi}{3}} \sin x \, dx \right| + \left| \int_{\frac{4\pi}{3}}^{\frac{3\pi}{2}} \sqrt{3} \cos x \, dx \right|$

$= \left| \left[-\cos x \right]_{\pi}^{\frac{4\pi}{3}} \right| + \left| \left[\sqrt{3} \sin x \right]_{\frac{4\pi}{3}}^{\frac{3\pi}{2}} \right|$

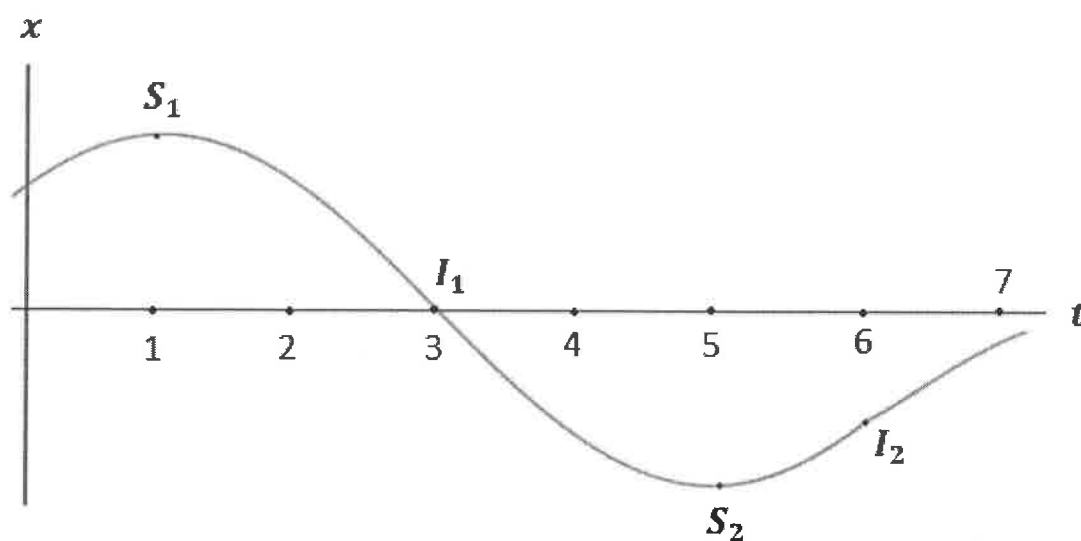
$= \left| -\cos \frac{4\pi}{3} - (-\cos \pi) \right| + \left| \sqrt{3} \sin \frac{3\pi}{2} - \sqrt{3} \sin \frac{4\pi}{3} \right|$

$= \left| \frac{1}{2} + (-1) \right| + \left| -\sqrt{3} - \sqrt{3} \left(-\frac{\sqrt{3}}{2} \right) \right|$

$= \frac{1}{2} + \left| -\sqrt{3} + \frac{3}{2} \right| = \frac{1}{2} + \sqrt{3} - \frac{3}{2} = (\sqrt{3} - 1) \text{ units}^2$

(as $\sqrt{3} > \frac{3}{2}$)

Question 27 (3 marks)



The graph shows the displacement of a particle, moving in a straight line, over the first 7 seconds of its motion. S_1 and S_2 are stationary points, and I_1 and I_2 are inflection points.

State the times, or periods of time, for which:

(a) The particle is stationary.

At $t=1$ and $t=5$

1

(b) The velocity is negative.

$$v = \frac{dx}{dt} \therefore$$

$1 < t < 5$

1

(c) The acceleration is positive.

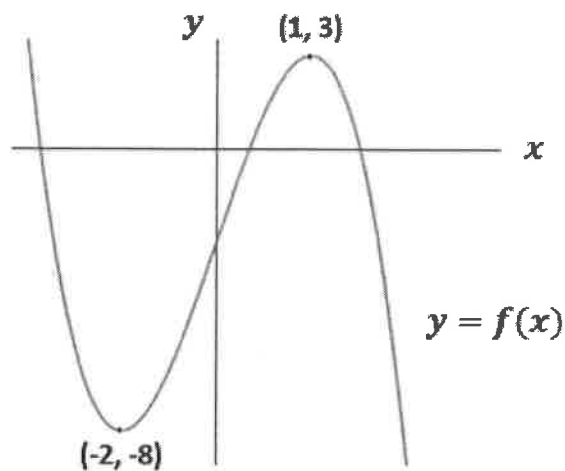
$$a = \frac{d^2x}{dt^2} \therefore \text{concave up} \therefore$$

$3 < t < 6$

1

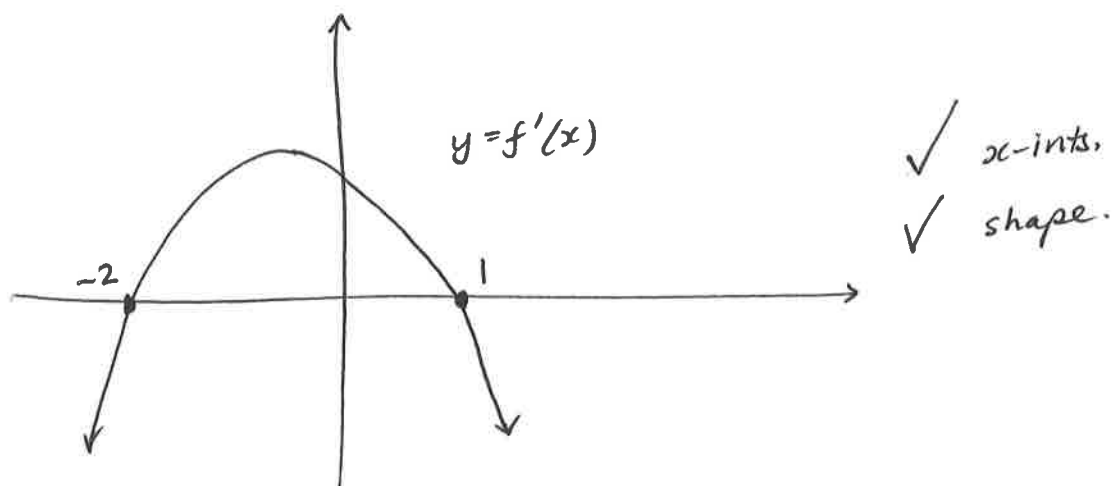
Question 28 (4 marks)

Consider the graph of $y = f(x)$ shown:



- (a) Use the space below to sketch the graph of $y = f'(x)$

2



- (b) Find the area bounded by $y = f'(x)$ and the x-axis.

2

$$\text{Area} = \int_{-2}^1 f'(x) \, dx$$

$$= [f(x)]_{-2}^1$$

$$= 3 - (-8) = 11$$

✓✓

Question 29 (4 marks)

For events A and B from a sample space, $P(A|B) = \frac{3}{4}$ and $P(B) = \frac{1}{3}$

(a) Calculate $P(A \cap B)$

1

$$\begin{aligned} P(A \cap B) &= P(A|B) \cdot P(B) \\ &= \frac{3}{4} \cdot \frac{1}{3} \\ &= \left(\frac{1}{4} \right) \end{aligned}$$

(b) Calculate $P(\bar{A} \cap B)$ where $\bar{A}$ denotes the complement of A.

1

$$\begin{aligned} P(\bar{A} \cap B) &= P(\bar{A}|B) \cdot P(B) \\ &= \frac{1}{4} \cdot \frac{1}{3} \\ &= \left(\frac{1}{12} \right) \end{aligned}$$

$$\begin{aligned} \text{[ALT: } P(\bar{A} \cap B) &= P(B) - P(A \cap B) \\ &= \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

(c) If A and B are independent, calculate $P(A \cup B)$

2

"Independent" means $P(A) = P(A|B) = \frac{3}{4}$

$$\begin{aligned} \therefore P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{3}{4} + \frac{1}{3} - \frac{1}{4} \\ &= \left(\frac{5}{6} \right) \end{aligned}$$

Question 30 (2 marks)

The gradient of a curve is given by $\frac{dy}{dx} = \frac{3x}{x^2+e}$

The curve passes through (0, 2). What is its equation?

2

$$\frac{dy}{dx} = \frac{3x}{x^2+e} = \frac{3}{2} \cdot \frac{2x}{x^2+e}$$

$$\therefore y = \frac{3}{2} \ln(x^2+e) + C$$

At $x=0, y=2$ $\therefore 2 = \frac{3}{2} \ln(0^2+e) + C$

$$2 = \frac{3}{2} \ln e + C$$

$$= \frac{3}{2} \cdot 1 + C \quad \therefore C = \frac{1}{2}$$

$$\therefore y = \frac{3}{2} \log_e(x^2+e) + \frac{1}{2}$$

Question 31 (3 marks)

If $f(x) = \sqrt{2-x}$ and $g(x) = \sqrt{x}$, then

(a) Find the rule for the composite function $f \circ g$

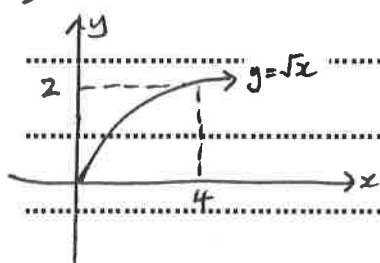
1

$$f \circ g(x) = \sqrt{2-\sqrt{x}}$$

(b) Find the domain of $f \circ g$

2

We need $x \geq 0$ (for $\sqrt{x}$) and also $2-\sqrt{x} \geq 0$.
ie $x \geq 0$ and $\sqrt{x} \leq 2 \quad \therefore x \leq 4$.



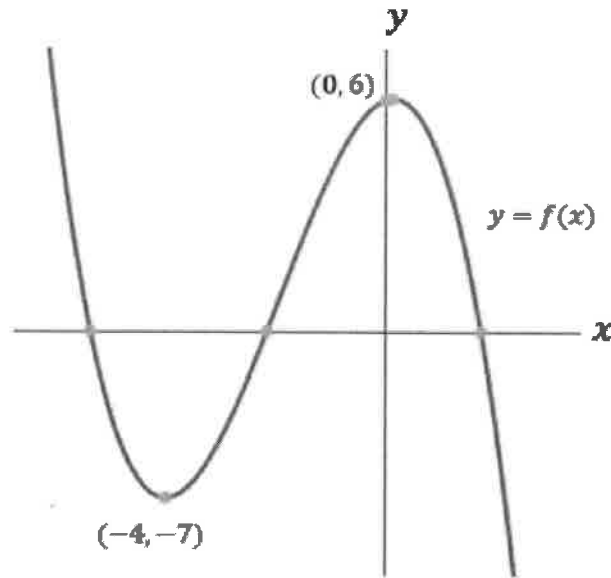
$$\therefore D: 0 \leq x \leq 4$$

$$(ie. [0, 4])$$

Question 32 (2 marks)

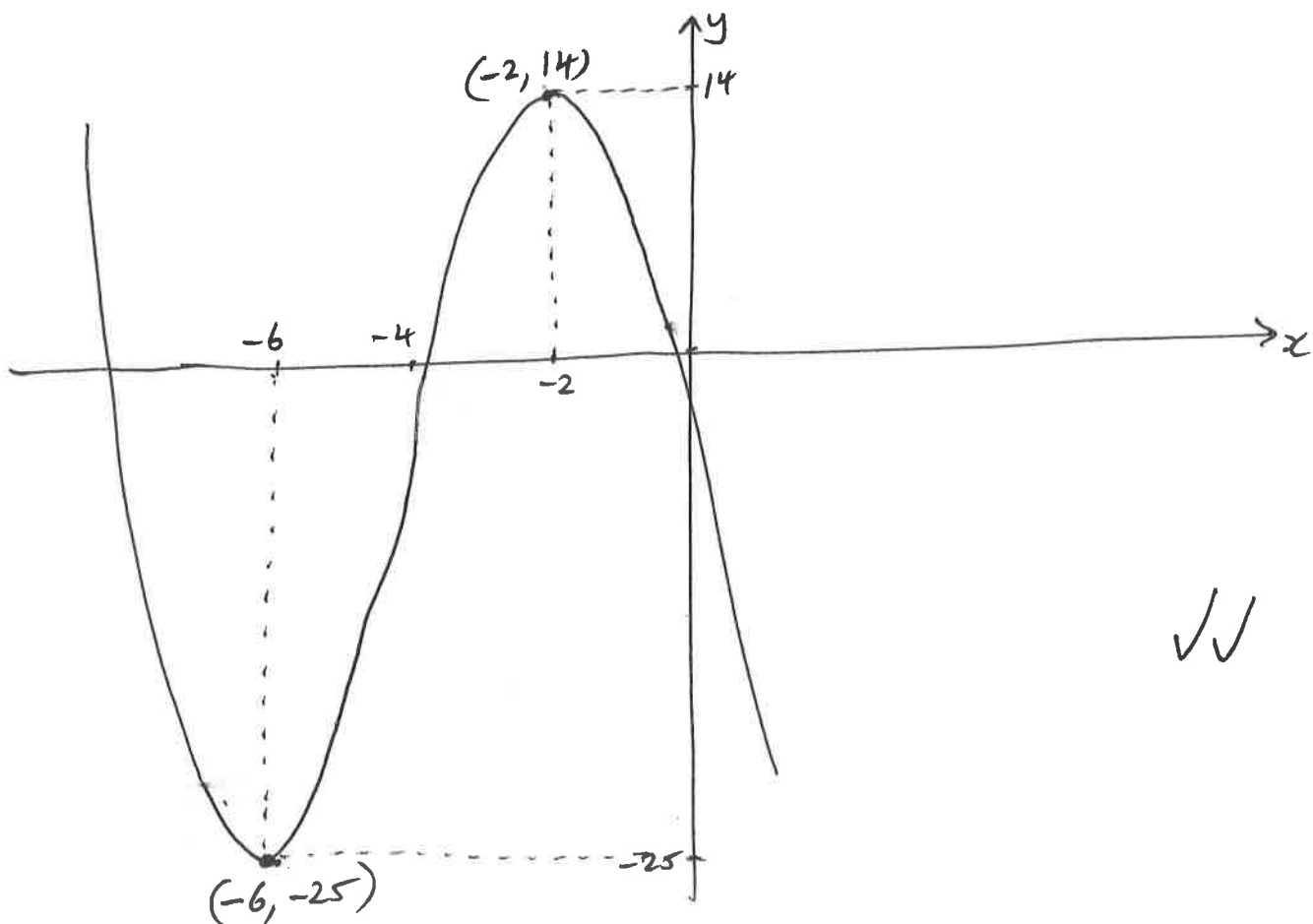
Given the graph of the function $y = f(x)$ below, with turning points as shown, sketch the transformed function $y = 3f(x+2) - 4$. (x-intercepts not required).

2



$(0, 6)$ transforms to $x = -2$
 $y = 3(6) - 4 = 14$ $\therefore (-2, 14)$

$(-4, -7)$ transforms to $x = -6$
 $y = 3(-7) - 4 = -25$ $\therefore (-6, -25)$.



Question 33 (5 marks)

- (a) Differentiate $y = \log_e (\cos x)$ with respect to x .

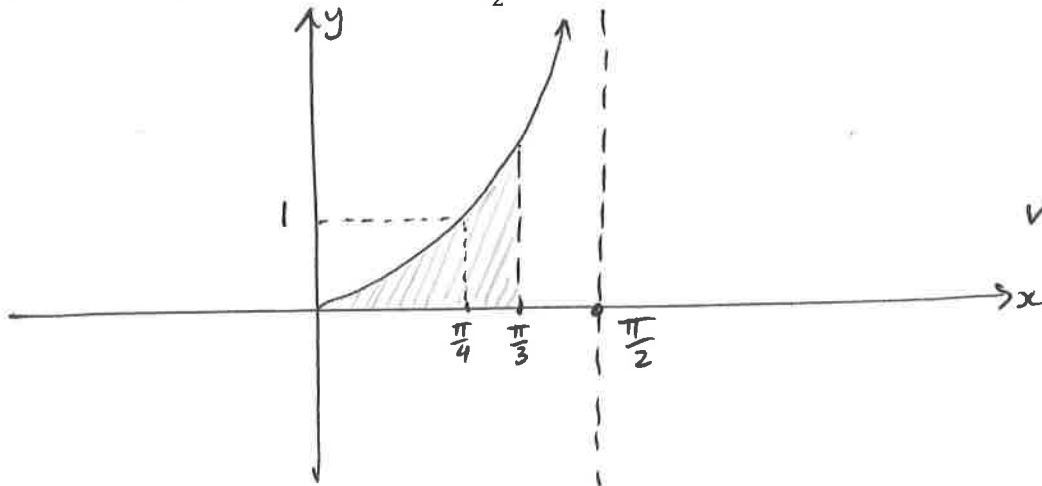
1

$$y' = -\sin x \cdot \frac{1}{\cos x}$$

$$= -\tan x$$

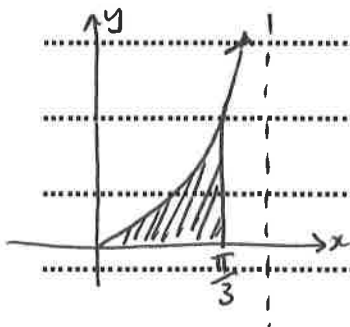
- (b) Sketch $y = \tan x$ for $0 \leq x \leq \frac{\pi}{2}$

1



- (c) Hence, using parts (a) and (b), find the area bounded by $y = \tan x$, the x -axis, and the line $x = \frac{\pi}{3}$ (leave answer in simplest exact form)

3



$$A = \int_0^{\pi/3} \tan x \, dx$$

$$= - \int_0^{\pi/3} -\tan x \, dx$$

$$= - \left[\log_e (\cos x) \right]_0^{\pi/3}$$

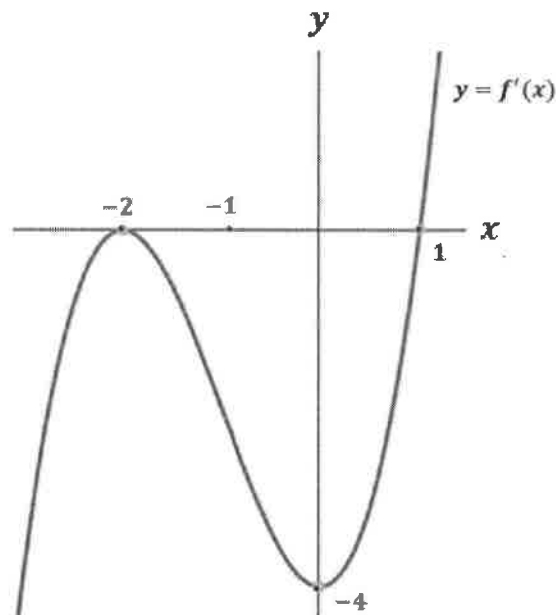
$$= - \left[\log_e \left(\frac{1}{2} \right) - \log_e 1 \right]$$

$$= - \left(\log_e 1 - \log_e 2 - \log_e 1 \right)$$

$$= \log_e 2 \text{ units}^2$$

Question 34 (3 marks)

The diagram shows $y = f'(x)$, the graph of the derivative function of $y = f(x)$.



- (a) Explain why there is a horizontal point of inflection at $x = -2$

1

At $x = -2$, $f'(x) = 0 \therefore$ it's a stationary point.

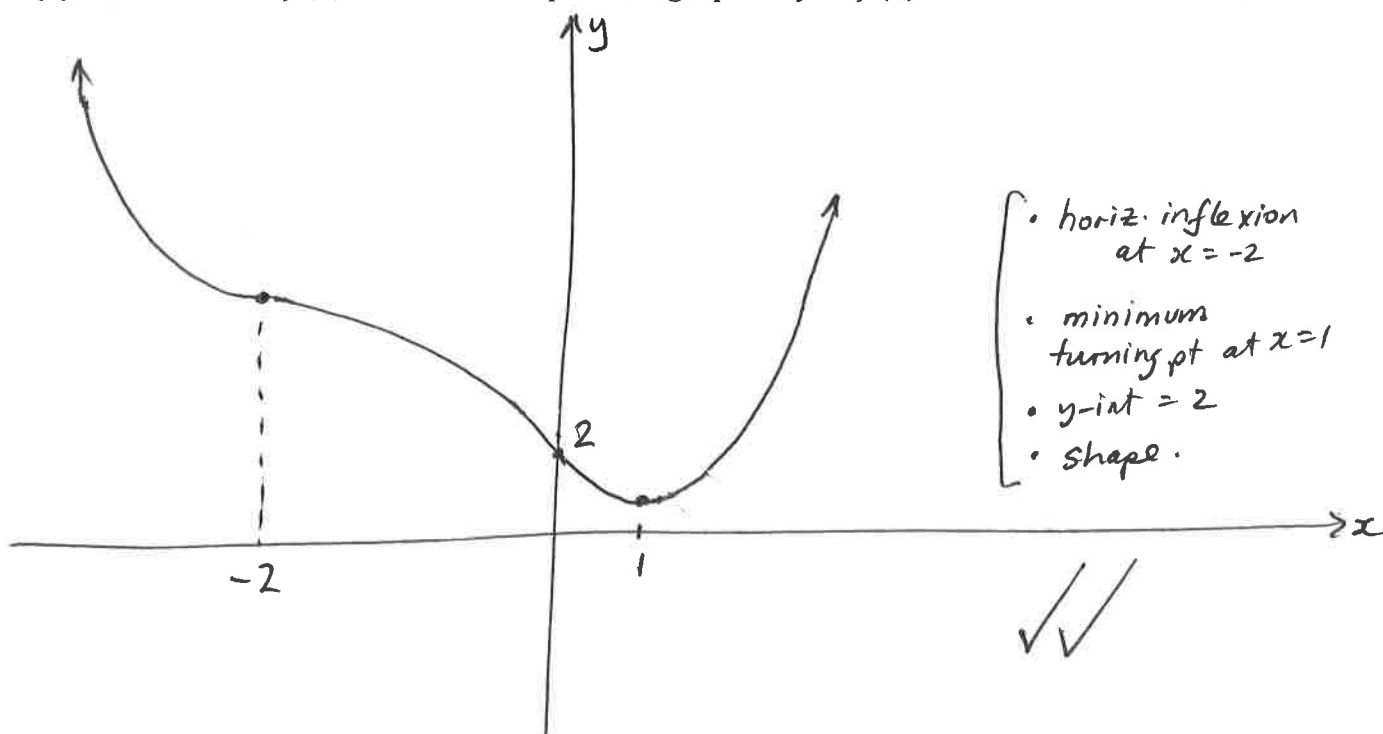
Also, $f''(x) > 0$ on left of $x = -2$, & $f''(x) < 0$ on right

$\therefore$ change in concavity $\therefore$ (horizontal) point of inflection.

(ALT: $f'(x)$ is 0 at $x = -2$ but negative on both sides of -2)
i.e. $\searrow$ $\therefore$ inflection.

- (b) Given that $f(0) = 2$, sketch a possible graph of $y = f(x)$.

2



Question 35 (3 marks)

Find the equation of the normal to $y = x \sin x$ at the point where $x = \frac{\pi}{2}$.

3

$$y = x \sin x \quad \therefore y' = \sin x \cdot 1 + x \cdot \cos x$$

$$\text{i.e. } y' = \sin x + x \cos x$$

$$\text{At } x = \frac{\pi}{2}, \quad y = \frac{\pi}{2} \cdot \sin \frac{\pi}{2} = \frac{\pi}{2}$$

$$\text{and, } y' = \sin \frac{\pi}{2} + \frac{\pi}{2} \cdot \cos \frac{\pi}{2} = 1 \quad \therefore m_N = -1$$

$$\therefore \text{normal is } y - \frac{\pi}{2} = -1 \left(x - \frac{\pi}{2} \right)$$

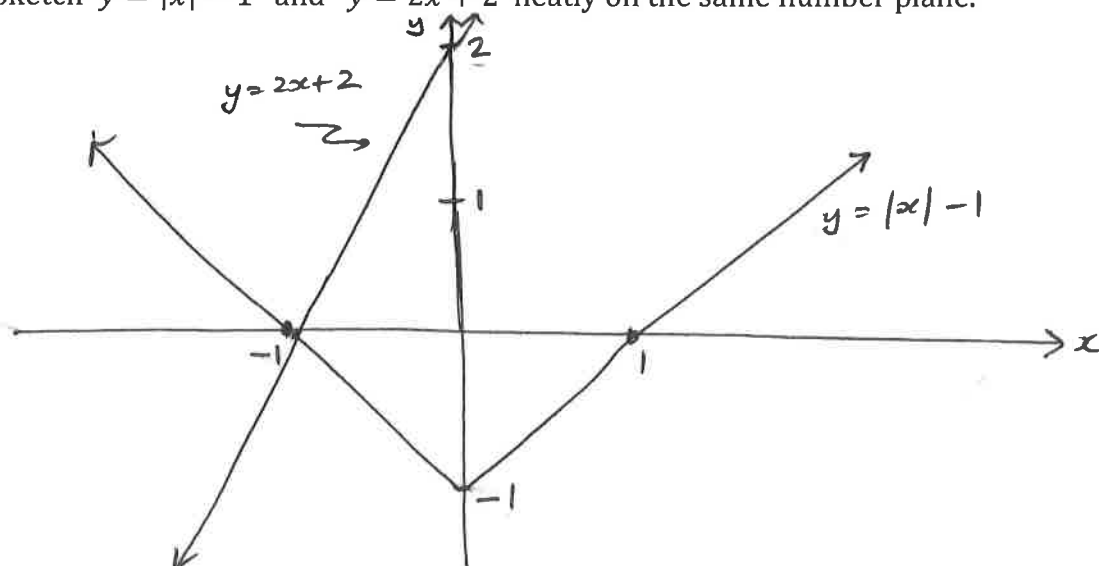
$$= -x + \frac{\pi}{2}$$

$$\therefore y = -x + \pi \quad (\text{i.e. } x + y - \pi = 0)$$

Question 36 (2 marks)

(a) Sketch $y = |x| - 1$ and $y = 2x + 2$ neatly on the same number plane.

1



(b) Hence solve the equation $|x| - 2x = 3$

1

$$\text{If } |x| - 2x = 3 \text{ then } |x| = 2x + 3$$

$$\therefore |x| - 1 = 2x + 2$$

So solution will be where graphs intersect in part (a)

$$\therefore x = -1$$

Question 37 (4 marks)

One half percent (0.5 %) of a country has a certain viral disease. A test is developed for the disease. The test gives a false positive 3% of the time, and a false negative 2% of the time.

- (a) Show that the probability that Andy, a randomly selected person, tests positive is 0.03475

2

[Hint: in this question, let D be the event that Andy has the disease, and $\bar{D}$ be the event Andy does not have it. Let T be the event that Andy's test comes back positive.]

$$\begin{aligned} P(T) &= P(T|D) \cdot P(D) + P(T|\bar{D}) \cdot P(\bar{D}) \\ &= (0.98)(0.005) + (0.03)(0.995) \\ &= 0.03475 \end{aligned}$$

✓✓

- (b) Andy just got the bad news that his test came back positive.

Find the probability that Andy actually has the disease.

2

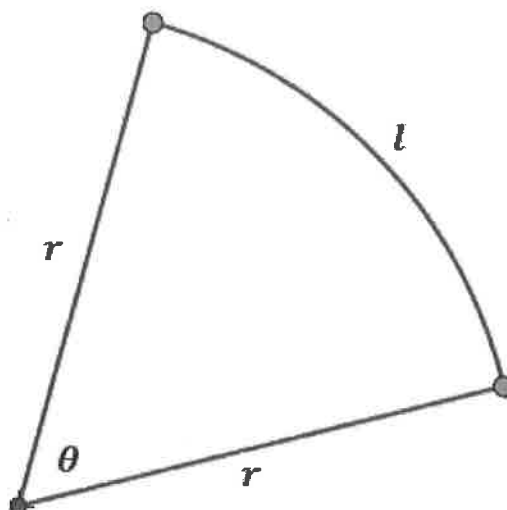
$$\begin{aligned} P(D|T) &= \frac{P(D \cap T)}{P(T)} = \frac{P(T \cap D)}{P(T)} \\ &= \frac{P(T|D) \cdot P(D)}{P(T)} \\ &= \frac{(0.98)(0.005)}{0.03475} \end{aligned}$$

$$\begin{aligned} &= 0.141 \dots \\ &\text{(i.e. } \hat{=} 14\%) \end{aligned}$$

✓✓

Question 38 (5 marks)

The diagram below shows a sector of a circle of radius r centimetres. The angle at the centre is θ radians, and the perimeter of the whole sector is 8 cm.



- (a) Show that $r = \frac{8}{2+\theta}$.

1

$$l = r\theta$$

$$\therefore r + r + r\theta = 8$$

$$2r + r\theta = 8$$

$$r(2+\theta) = 8 \quad \therefore r = \frac{8}{2+\theta}, \text{ as required.}$$

✓

- (b) Show that A , the area of the sector in cm^2 , is given by

$$A = \frac{32\theta}{(\theta+2)^2}$$

1

$$A = \frac{1}{2}r^2\theta = \frac{1}{2}\left(\frac{8}{2+\theta}\right)^2 \cdot \theta = \frac{1}{2} \cdot \frac{64\theta}{(2+\theta)^2}$$

$$\therefore A = \frac{32\theta}{(2+\theta)^2}, \text{ as required.}$$

✓

- (c) If $0 \leq \theta \leq \frac{\pi}{2}$, find the maximum area of the sector, and the value of θ for which this occurs.

3

$$\frac{dA}{d\theta} = \frac{(\theta+2)^2 \cdot 32 - (32\theta) \cdot 2(\theta+2)}{(\theta+2)^4}$$

✓

For max. A, $\frac{dA}{d\theta} = 0 \quad \therefore 32(\theta+2)^2 - 64\theta(\theta+2) = 0$

$$\therefore 32(\theta+2)[\theta+2 - 2\theta] = 0$$

$$\therefore 32(\theta+2)(2-\theta) = 0$$

$$\therefore \theta = -2 \text{ or } \theta = 2$$

But $0 \leq \theta \leq \frac{\pi}{2}$, so these both lie outside the possible range of θ .

So we need to test the endpoints, 0 and $\frac{\pi}{2}$:-

At $\theta = 0$, $A = 0$

$$\text{At } \theta = \frac{\pi}{2}, A = \frac{32\left(\frac{\pi}{2}\right)}{\left(\frac{\pi}{2} + 2\right)^2} = \frac{16\pi}{\left(\frac{\pi+4}{2}\right)^2}$$

$$\therefore A = \frac{64\pi}{(\pi+4)^2}$$

$$\doteq 3.9422 \text{ cm}^2$$

Thus max. area = $\frac{64\pi}{(\pi+4)^2} \text{ cm}^2 \doteq 3.9422 \text{ cm}^2$,
obtained when $\theta = \frac{\pi}{2}$ radians.

✓✓

End of examination